

Optics

↳ Branch of physics which deals about the study of light & its associated phenomena, is called Optics.

• Types of optics

1. Geometrical Optics
2. Physical Optics
3. Quantum Optics

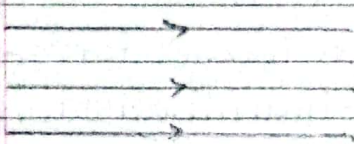
★ Some Terminologies in Optics

• Ray

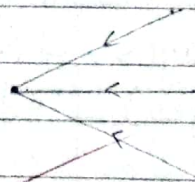
↳ A straight line representing light's path

• Beam

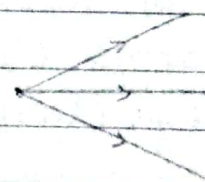
↳ A group of rays is Beam.



- Parallel Beam



- Convergent Beam

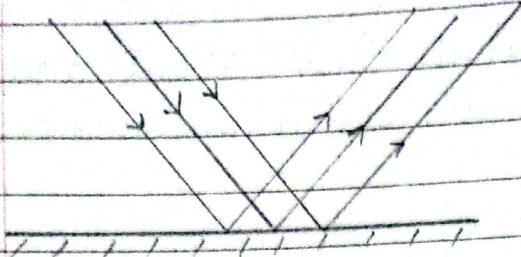


- Divergent Beam

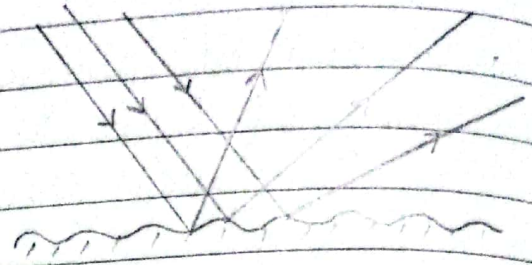
• Reflection

↳ The process of returning of light on the same medium after striking on the surface is called reflection of light.

• Types of Reflection



- Regular Reflection



- Irregular Reflection

• Difference betⁿ Regular Reflection and Irregular Reflection

Regular Reflection	Irregular Reflection
↳ Occurs on smooth, polished surfaces. (e.g - Mirrors)	↳ Occurs on Rough Surfaces (e.g - Paper, wall)
↳ Light rays reflect parallel to each other.	↳ Light rays scatter in random directions
↳ forms clear images (e.g: Mirror images)	↳ No clear image formed, only illumination.
↳ e.g: Reflection in a plane mirror.	↳ e.g: light reflecting off a book page.

• Laws of Reflection of Light

- ↳ The Angle of Incidence (i) is equal to the Angle of Reflection (r).
i.e. $i = r$
- ↳ Incident ray, reflected ray and Normal all lie on the same point of same plane.

• Image formation

- ↳ An image is the visual reproduction of an object formed when light rays either actually meet or appear to meet after reflection or refraction.

• Types of Images

1. Real Image

- ↳ It is formed by actual intersection of light rays.
- ↳ It can be projected on a screen (e.g. image on a camera sensor.)
- ↳ Inverted with respect to object (upside-down)
- ↳ Example: Image formed by a convex lens or concave mirror. (in general)

2. Virtual Image

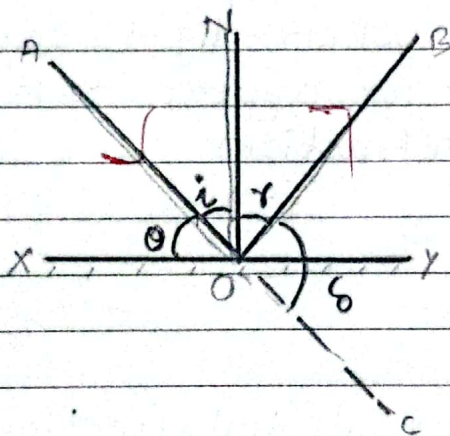
↳ It is formed when light ray appears to meet at a point.

↳ It cannot be projected on a screen

↳ Upright / erect (same orientation as the object)

↳ example: Image seen in a plane mirror.

• Deviation Produced by plane mirror:



In figure,

XY = Plane mirror

ON = Normal

AO = Incident ray

OB = Reflected ray

OC = Original direction of incident ray

$\angle AON = i =$ incident angle

$\angle BON = r =$ reflected angle

$\angle BOC = \delta =$ $\angle$ of deviation

$\angle AOX = \theta =$ glancing angle

- Glancing angle (θ)

↳ The angle made by incident ray with a reflecting surface or refracting surface is called glancing angle.

- Angle of deviation (δ)

↳ The angle made by reflected ray with the original direction of incident ray is called angle of deviation.

In figure,

$\angle BOC = \delta$ is the angle of deviation.

Now, From figure

$$\angle BOC = \angle BOY + \angle COY$$

$$\delta = (90 - r) + \theta \quad [\because \angle AOX = \angle COY]$$

$$\delta = 90 - i + \theta \quad [\because i = r]$$

$$\delta = \theta + \theta$$

$$\boxed{\delta = 2\theta}$$

Hence, the angle of deviation is twice that of the glancing angle.

- Nature of image formed by plane mirror

↳ Virtual (For real object)

↳ Same sized

↳ Object distance = image distance

↳ Laterally inverted (left-right inversion)

↳ Erect

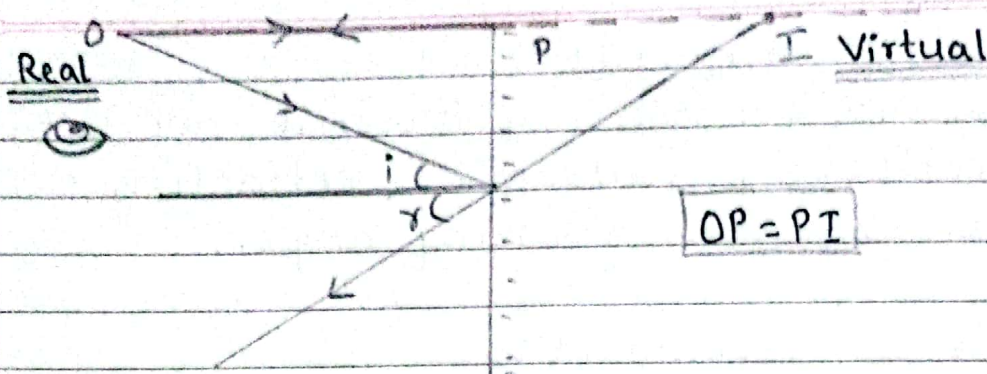


fig: Image formation by plane mirror.

• Notes on plane mirror

1. Relative speed on Image in a plane mirror

→ When an object moves towards or away from a plane mirror with speed u , the image moves with speed u relative to the mirror.

→ The relative speed of the image with respect to the object is $2u$.

2. Mirror moving towards a stationary object

→ If the mirror moves towards a stationary object with speed u , the image moves with speed $2u$.

3. Minimum mirror height for full-length image

→ A person of height ' h ' requires a mirror of at least $h/2$ length to see their complete image.

4. Mirror ~~Height~~ Height for viewing a wall behind

↳ To see the entire height of a wall behind them, a person standing in the middle of the room needs a mirror with a height of at least $\frac{1}{3}$ the height of the wall.

5. Images by two inclined plane mirrors

↳ When two plane mirrors are inclined to each other at an angle θ , then number of images (n) formed of an object which is kept between them. $n = \left(\frac{360}{\theta} - 1 \right)$, for

object placed Symmetrically.

6. Law of rotation of light

↳ It states that when a mirror rotates through an angle θ , then the reflected ray rotates through angle 2θ .

Spherical (curved) mirror

↳ It is the part of hollow spherical reflecting surface.

① Concave mirror (converging mirror)

↳ It is the spherical mirror with inner part reflecting.

↳ This mirror is also called converging

mirror because it converges parallel ray of light at a point after reflection.

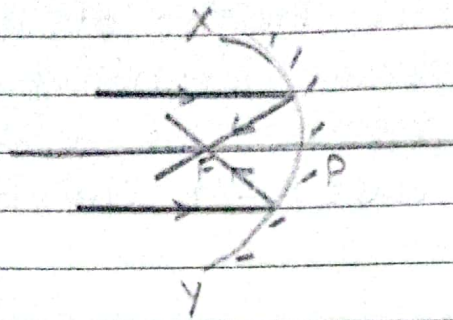


fig - Concave mirror

② Convex mirror (diverging mirror)

↳ It is the spherical mirror with outer part reflecting.

↳ This mirror is also called diverging mirror because it diverges parallel ray of light after reflection.

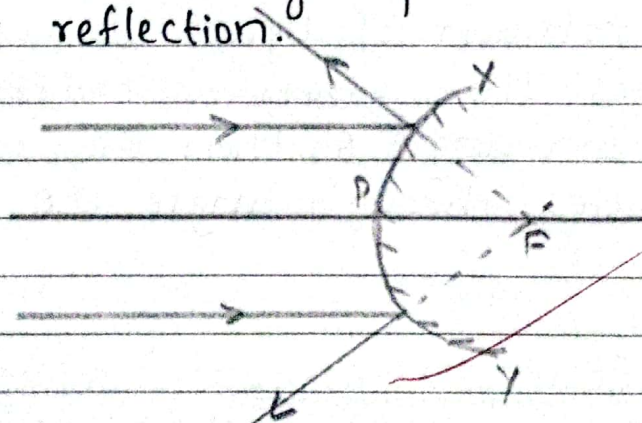
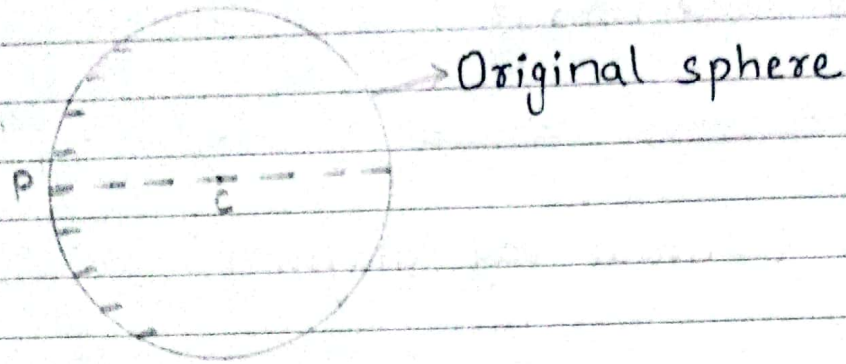


fig - Convex mirror

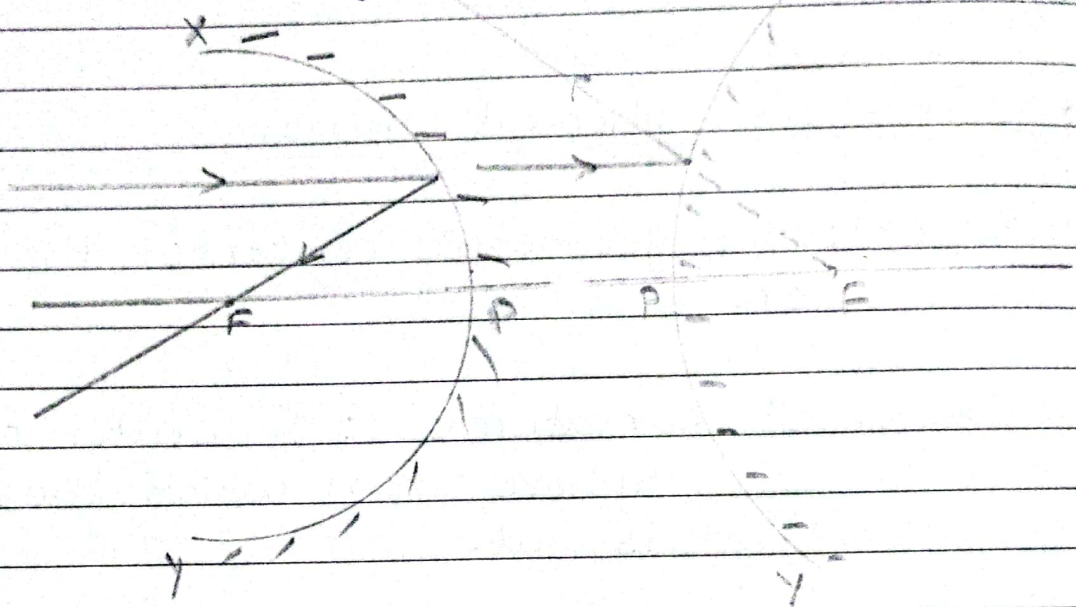


• Terms in spherical mirror

1. Pole (P): The geometric center of the mirror.
or.
2. Center of curvature (C): The center of the sphere from which the mirror is cut.
3. Radius of curvature (R): Distance between P and C
4. Principal Axis: The straight line passing through P and C.
5. Focus (F): The point where parallel rays converge (concave mirror) or appear to diverge from (convex mirror)
6. Focal length (f): Distance between P and F.
7. Aperture: It is the geometrical size of the mirror (XY).

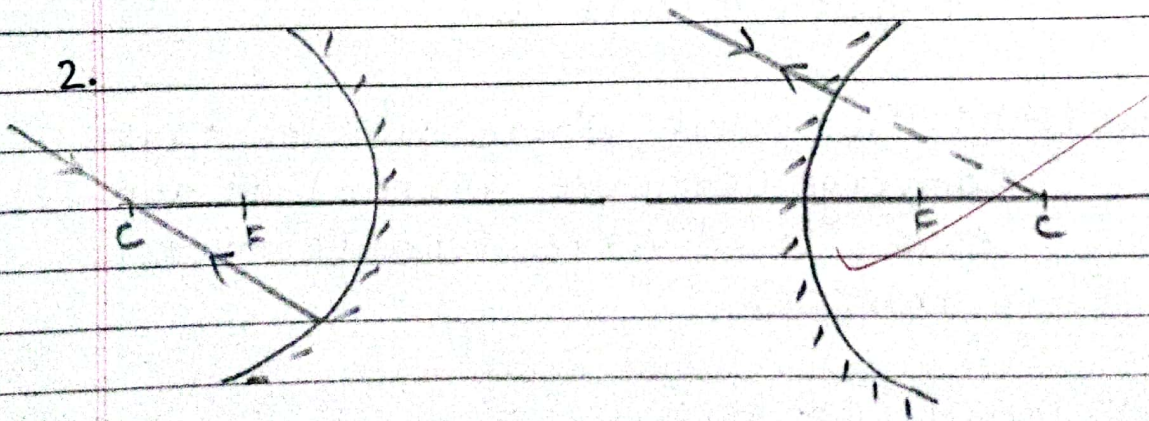
Rules to draw ray diagram

1.



→ 1. A Ray of light parallel to principal axis passes through the focus or appears to come from the focus after reflection and its converse is also true.

2.



→ A ray of light from center of curvature or directed towards the center of curvature, returns from the same path

after reflection.

Prove that : $f = \frac{R}{2}$

① For concave mirror

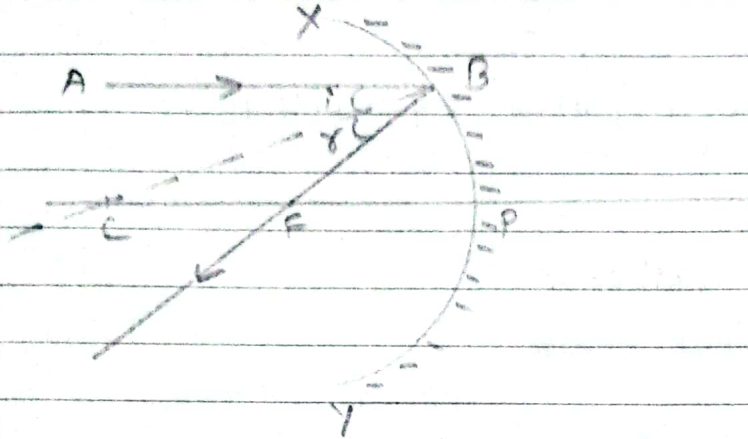


fig- Reflection from concave mirror

Consider a concave mirror XY of focal length 'f' radius of curvature 'R'. AB is the incident ray parallel to principal axis which passes through the focus after reflection as shown in figure. CB is the perpendicular line at Point B. So, angle $\angle ABC = i$ is the incident angle and $\angle CBF = r$ is the reflected angle.

Now, From figure,

1) $\angle BCF = i$ (Alternate angle)

2) $\angle CBF = \angle BCF$ [$\because i = r$]

3) $\triangle BCF$ is the isosceles triangle
 $\therefore CF = BF$

4) For small aperture of the mirror
B and P are very close to each other.

$$\therefore BF \approx PF$$

Now,

$$PC = PF + CF$$

$$PC = 2PF$$

$$\boxed{R = 2f} \text{ or } \boxed{f = \frac{R}{2}}$$

This is the required equation.

② For Convex mirror

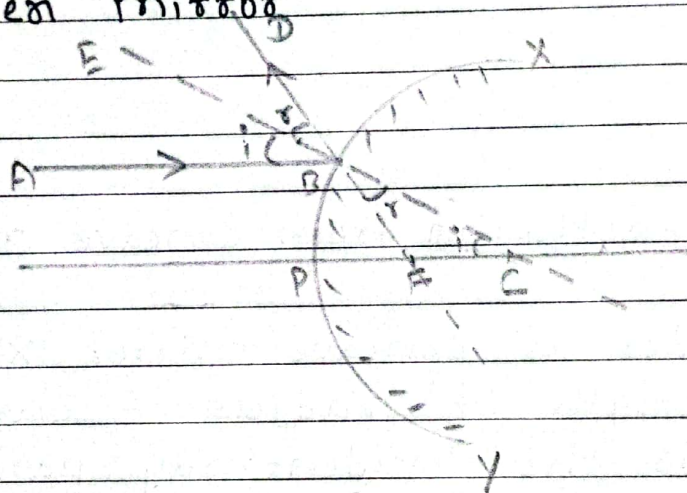


fig: Reflection from convex mirror

Consider a convex mirror XY of focal length 'f' and radius of curvature 'R'. AB is the incident ray parallel to principal axis which appears to come from the focus after reflection as shown in figure. EC is the perpendicular line of point B. So, $\angle ABE = i$ is the angle of incidence and $\angle DBE = r$ is the angle of reflection.

In figure,

1) $\angle CBF = r$ [Vertically Opposite $\angle$]

2) $\angle FCB = i$ (Corresponding angle)

3) $\triangle BCF$ is the isosceles triangle
[$\because i = r$]

$\therefore BF = CF$

4) For small aperture of mirror
B & P are very close to each other

$\therefore BF \approx PF$

Now,

$$PC = PF + CF$$

$$PC = 2PF$$

$$\boxed{R = 2f} \text{ or } \boxed{f = \frac{R}{2}}$$

This is the required equation.

Mirror Formula

↳ Mirror Formula is the relationship between object distance (u), Image distance (v) and focal length (f) of the spherical mirror.

$$\text{i.e.} - \left[\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \right] \underline{\text{OR}} \left[\frac{2}{R} = \frac{1}{u} + \frac{1}{v} \right]$$

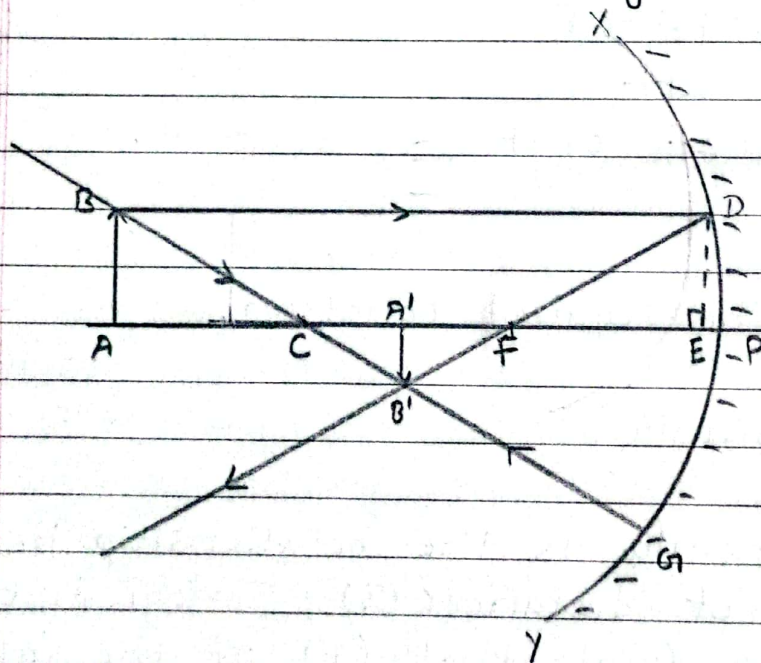
Assumptions & sign convention.

1. The small aperture of mirror is taken.

2. Focal length of concave mirror is taken as positive & that of convex mirror is taken as negative.

3. Object and image distance for real is taken as +ve and for virtual it is taken as -ve.

• Mirror formula for concave mirror when real ~~is~~ ~~formed~~ image is formed:



Consider a concave mirror XY of focal length ' f ' and Radius of curvature ' R '. AB is the object on principal axis beyond C. A incident ray BD parallel to principal axis, passes through the focus after reflection. Another Ray BG from center of curvature returns from the same path after ref-

lection. So, $A'B'$ is the real and inverted image in between C and F . as shown in figure.

Draw: DE perpendicular to CP .

Now, From figure

$$\triangle ABC \sim \triangle A'B'C \text{ [A.A.A fact]}$$

$$\therefore \frac{AB}{A'B'} = \frac{AC}{A'C} \rightarrow (i)$$

Again,

$$\triangle DEF \sim \triangle A'B'F \text{ [by A.A.A fact]}$$

$$\therefore \frac{DE}{A'B'} = \frac{EF}{A'F}$$

$$\Rightarrow \frac{AB}{A'B'} = \frac{EF}{A'F} \text{ [}\because AB = DE\text{]}$$

For the small aperture of mirror E and P are very close

$$\therefore EF = PF$$

$$\therefore \frac{AB}{A'B'} = \frac{PF}{A'F} \rightarrow (ii)$$

From eqⁿ (i) & (ii)

$$\frac{AC}{A'C} = \frac{PF}{A'F}$$

$$\Rightarrow \frac{PA - PC}{PC - PA'} = \frac{PF}{PA' - PF} \rightarrow (iii)$$

Where,

$PA = u$ (Object distance)

$PC = R \neq 2f$ (Radius of curvature)

$PA' = v$ (image distance)

$PF = f$ (focal length)

Using these values in eqn (iii)

$$\frac{u-2f}{2f-v} = \frac{f}{v-f}$$

$$\text{or, } uv - uf - 2vf + 2f^2 = 2f^2 - vf$$

$$\text{or, } uv - uf = 2vf - vf$$

$$\text{or, } uv - uf = vf$$

$$\text{or, } uf = uv - vf$$

$$\text{or, } \frac{uf}{uvf} = \frac{uv}{uvf} - \frac{vf}{uvf}$$

$$\text{or, } \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

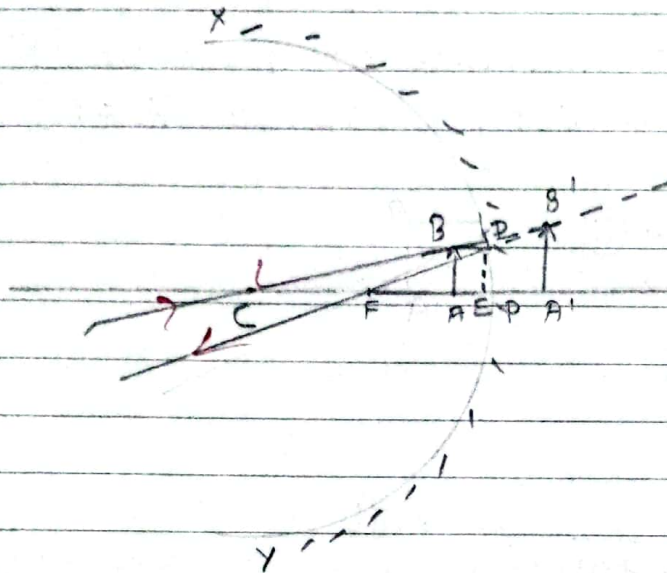
$$\therefore \frac{1}{f} = \frac{1}{u} + \frac{1}{v} \quad \therefore \text{This is the required mirror formula.}$$

Also,

$$f = \frac{R}{2}$$

$$\Rightarrow \frac{2}{R} = \frac{1}{u} + \frac{1}{v}$$

- Mirror formula using concave mirror:
When virtual image is formed.



Consider a concave mirror XY of focal length ' f ' and Radius of curvature ' R '. AB is the object on principal axis in between P and F . So, $A'B'$ is the virtual and erect image of AB behind the mirror as shown in Figure. Draw DE perpendicular to CP .

Now, From Figure,

$$\triangle ABC \sim \triangle A'B'C \text{ [by AAA axiom]}$$

$$\therefore \frac{AB}{A'B'} = \frac{AC}{A'C} \quad \text{--- (1)}$$

Again,

$$\triangle DEF \sim \triangle A'B'F \text{ [by AAA axiom]}$$

$$\therefore \frac{DE}{A'B'} = \frac{EF}{A'F} \quad \text{--- (2)}$$

$$\Rightarrow \frac{AB}{A'B'} = \frac{PF}{A'F} \rightarrow (ii)$$

[$\therefore$ For small aperture of mirror F and P are very close so, $F \approx P$]

From eqn (i) & (ii)

$$\frac{AC}{A'C} = \frac{PF}{A'F}$$

$$\text{or, } \frac{PC - PA}{PC + PA'} = \frac{PF}{PF + PA'} \rightarrow (iii)$$

Where $PC = R = 2f = \text{Radius of curvature}$

$PA = u = \text{Object distance}$

$PA' = -v = \text{image distance (virtual)}$

$PF = f = \text{focal length}$

Using these values on eqn (iii)

$$\frac{2f - u}{2f + (-v)} = \frac{f}{f + (-v)}$$

$$\text{or, } \frac{2f - u}{2f - v} = \frac{f}{f - v}$$

$$\text{or, } 2f^2 - uf - 2vf + uv = 2f^2 - vf$$

$$\text{or, } uv + vf = 2vf + uf$$

$$\text{or, } uv = vf + uf$$

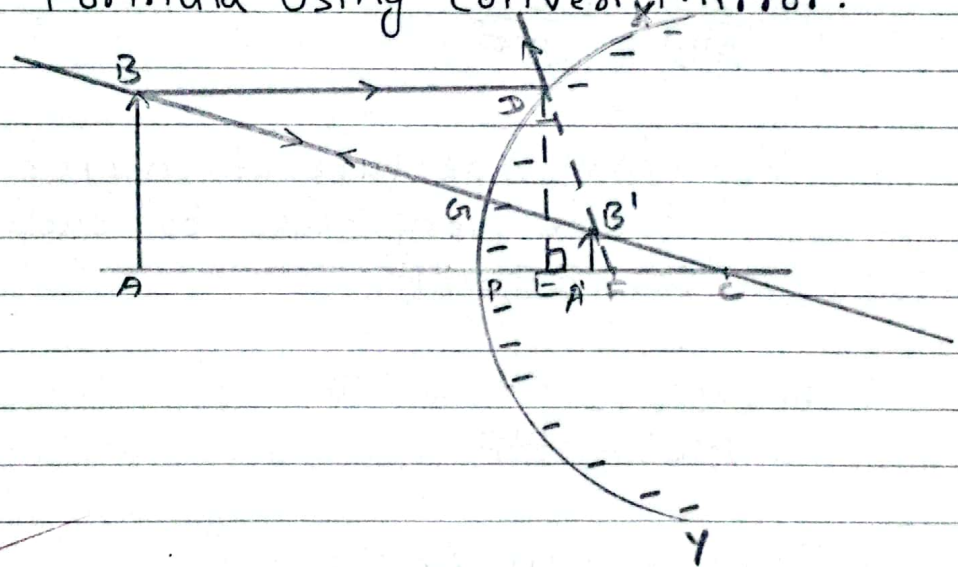
or, ~~2~~ $UV = f(u+v)$

or, $\frac{1}{f} = \frac{u+v}{uv}$

or, $\frac{1}{f} = \frac{u}{uv} + \frac{v}{uv}$

or, $\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$
Proved.

③ Mirror Formula Using convex mirror.



Consider a convex mirror XY of focal length 'f' and Radius of Curvature 'R'. AB is the real object on principal axis in ~~finite~~ finite point. So, A'B' is the virtual and erect image of AB behind the mirror as shown in figure. Draw DE perpendicular to AF.

Now,

From Figure,

$\triangle ABC \sim \triangle A'B'C$ [By A.A.A axiom]

$$\frac{AB}{A'B'} = \frac{AC}{A'C} \rightarrow \textcircled{i}$$

Again,

$\triangle DEF \sim \triangle A'B'F$

$$\frac{DE}{A'B'} = \frac{EF}{A'F}$$

$$\Rightarrow \frac{AB}{A'B'} = \frac{EF}{A'F}$$

For small aperture of mirror $\angle E$ and P are very close to each other,
so, $EF = PF$

$$\Rightarrow \frac{AB}{A'B'} = \frac{PF}{A'F} \rightarrow \textcircled{ii}$$

From eqn (i) & (ii)

$$\frac{AC}{A'C} = \frac{PF}{A'F} \rightarrow \textcircled{iii}$$

Where,

$\angle PF = -f$ (focal length - virtual)

$PA' = -v$

$PC = -2f$

$PA = u$

Using these values on eqn (iii)

$$\frac{AP + PC}{PC - PA'} = \frac{PF}{PF - PA'}$$

$$\text{or, } \frac{u + (-2f)}{-2f + v} = \frac{-f}{-f + v}$$

$$\text{or, } -uf + 2f^2 + uv - 2vf = 2f^2 - vf$$

$$\text{or, } uf - vf = uv - 2vf$$

$$\text{or, } uf + vf = uv$$

$$\text{or, } f \frac{(u+v)}{uv} = 1$$

$$\text{or, } \frac{1}{f} = \frac{u}{uv} + \frac{v}{uv}$$

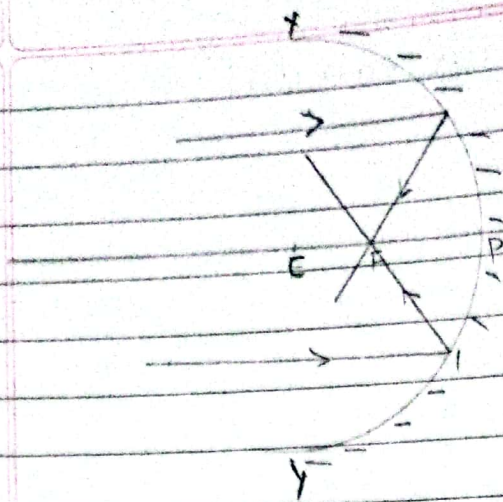
$$\therefore \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Nature of Image formed by spherical mirror.

① Concave

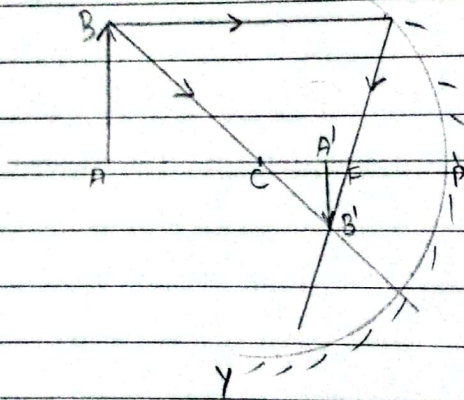
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② Object at infinity



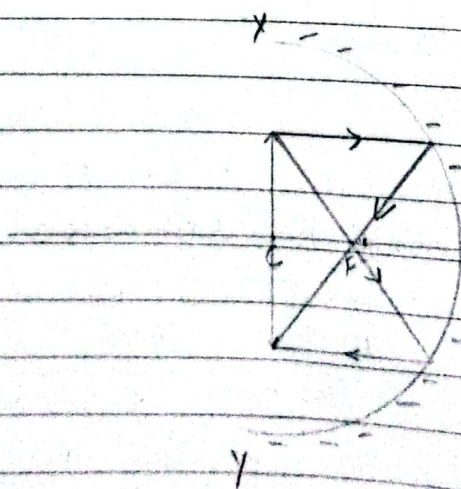
- Real
- Inverted
- Highly diminished
- Formed at F

② Object beyond C



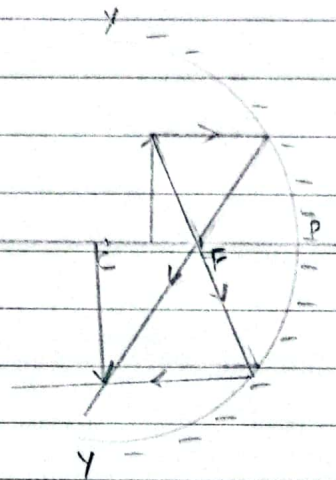
- Real
- Inverted
- Formed in between C and F
- diminished

③ Object at C



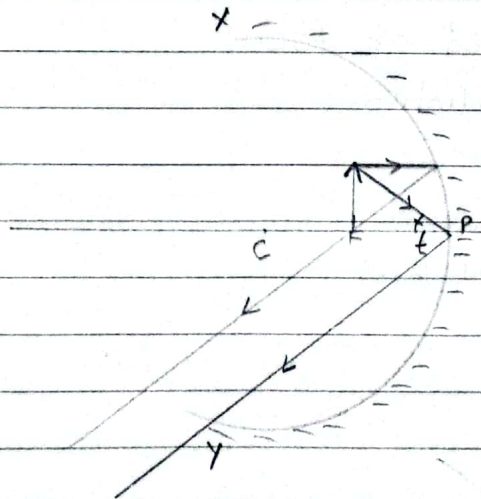
- Real
- Inverted
- Same Sized
- Formed at C.

(iv) Object between C and F



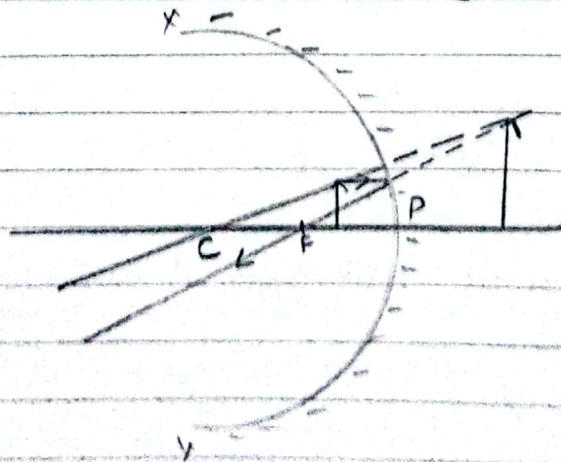
- Real
- Inverted
- Enlarged
- Image formed beyond C.

(v) Object at F.



- Real
- Inverted
- Highly Magnified
- Image formed at infinity (∞)

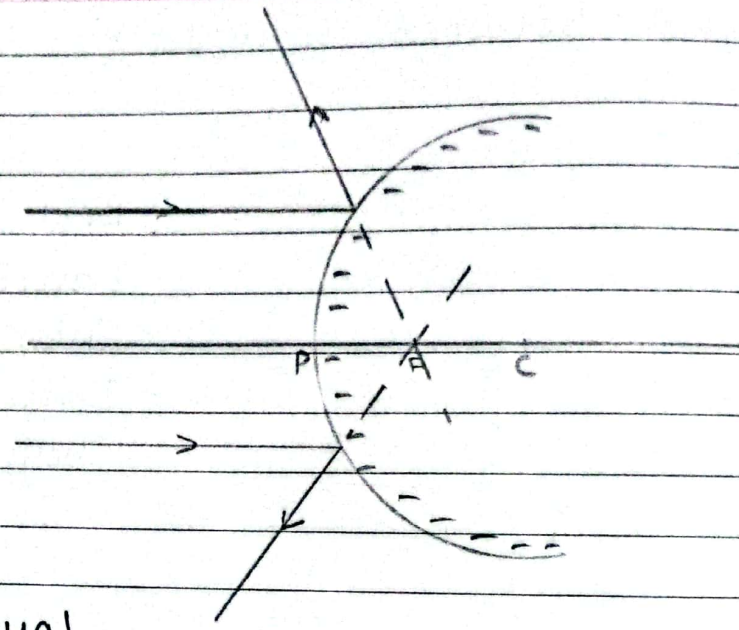
(vi) Object in between F and P.



- Virtual
- Erect
- Magnified
- Image formed behind the mirror

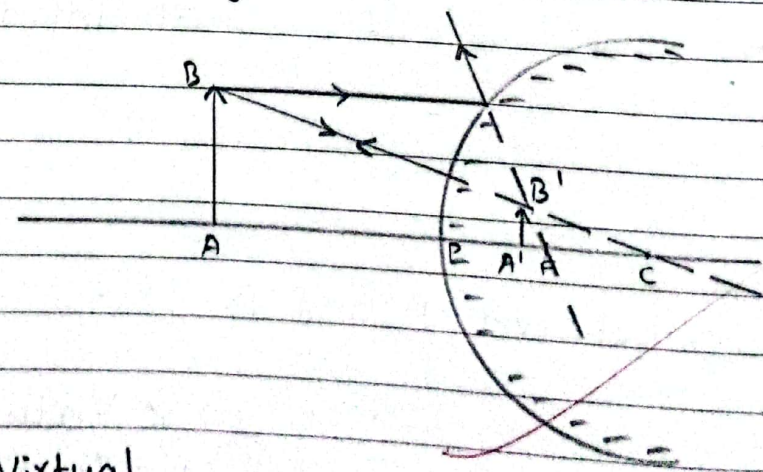
② Conver

① At infinity



- Virtual
- erect
- image formed at focus
- highly diminished

(ii) between infinity and pole



- Virtual
- erect
- Image formed between P and F
- diminished

Magnification (m):

→ Magnification is the ratio of size of the image to the size of the object.

i.e: $m = \frac{\text{Size of image (I)}}{\text{Size of object (O)}}$

⇒ $m = \frac{I}{O}$

→ It has no unit

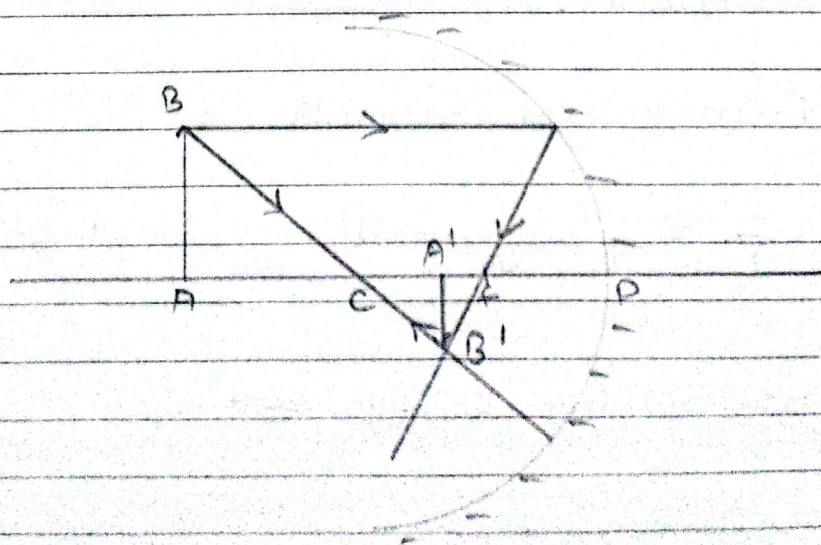
→ If $m = 1$, object and image have same size.

→ If $m > 1$, image is greater than object

→ If $m < 1$, image is less than object

→ If $m = -ve$, either image or object is virtual.

Magnification using Concave mirror.



Consider a concave mirror XY of focal length 'f' and Radius of curvature 'R'. AB is the real object on principle axis so, A'B' is the real image shown in figure.

From figure,

$\triangle ABP \sim \triangle A'B'P$ [by A.A.A axiom]

$$\therefore \frac{AB}{A'B'} = \frac{AP}{A'P}$$

$$\Rightarrow \frac{O}{I} = \frac{u}{v}$$

$$\Rightarrow \frac{I}{O} = \frac{v}{u}$$

$$\Rightarrow m = \frac{v}{u}$$

Hence,

Magnification can also be defined as the ratio of image distance to object distance.

From mirror Formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \rightarrow \textcircled{1}$$

multiplying $\textcircled{1}$ by ~~eq~~ v

$$\frac{v}{f} = \frac{v}{u} + \frac{v}{v}$$

$$\Rightarrow \frac{v}{f} = m+1$$

$$\Rightarrow m = \frac{v}{f} - 1$$

$$\Rightarrow m = \frac{v-f}{f}$$

Again, Multiplying eqⁿ(i) by u

$$\frac{u}{f} = \frac{u}{u} + \frac{u}{v}$$

$$\Rightarrow \frac{u}{f} = 1 + \frac{1}{m}$$

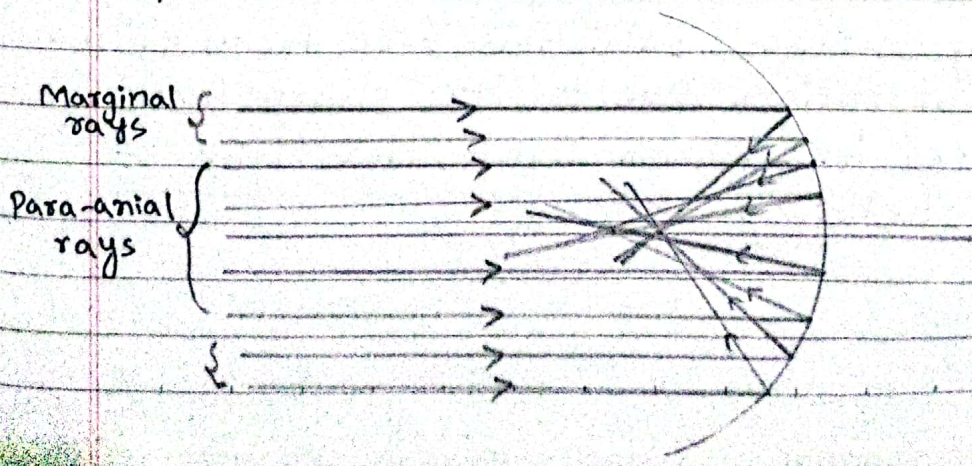
$$\Rightarrow \frac{1}{m} = \frac{u}{f} - 1$$

$$\Rightarrow \frac{1}{m} = \frac{u-f}{f}$$

$$\Rightarrow m = \frac{f}{u-f}$$

$$\therefore m = \frac{T}{O} = \frac{v}{u} = \frac{v-f}{f} = \frac{f}{u-f}$$

Spherical aberration



→ The inability of spherical mirror to focus marginal rays and para-axial rays at a point after reflection is called spherical ~~aberration~~ aberration.

- Method to remove spherical aberration.
- ① By using small aperture of spherical mirror.
 - ② By using parabolic mirror.

Parabolic Mirror

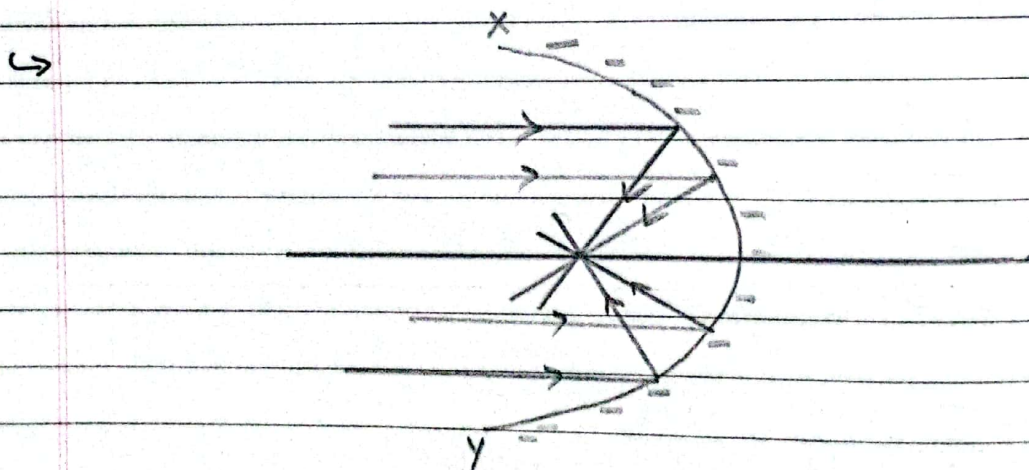


fig- Parabolic Mirror

→ Parabolic Mirror is the part of parabolic reflecting surface which is free from spherical aberration. Due to its sharp focus it is used in search light, headlight of vehicles, telescope etc.